REQUIREMENT SPECIFICATION DOCUMENT

E-COMMERCE ORDER MANAGEMENT SYSTEM

1. Introduction

1.1 Purpose

The purpose of this Requirement Specification Document is to analyze and document the business requirements of an **E-Commerce Order Management System**.

The system is a database-driven application designed to manage customers, products, sellers, orders, payments, shipping, and order status. It provides centralized storage and helps customers place and track orders while allowing administrators and business users to manage the complete order lifecycle.

1.2 System Overview

The E-Commerce Order Management System manages online product ordering activities. Customers can browse products, select products, place orders, make payments, and track deliveries. Administrators can manage customers, sellers, products, orders, payments, and reports.

1.3 Business Environment

E-commerce businesses handle large numbers of transactions. Manual or poorly organized processes may cause data duplication, incorrect orders, payment issues, inventory inconsistencies, delayed reports, and difficulty tracking orders. A centralized relational database helps overcome these problems.

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2. Business Problem, Objectives and Scope

2.1 Problem Statement

The business requires a system that can efficiently manage:

- Customer and seller information
- Product details and prices
- Customer orders
- Multiple products within an order
- Order amount calculation
- Payment information
- Shipping and delivery
- Order tracking and history
- Business reports

2.2 Objectives

The main objectives are:

1. Maintain accurate customer information.
2. Manage sellers and their products.
3. Maintain product price and availability.
4. Create, modify, cancel, and track orders.
5. Manage multiple products in an order.
6. Record payment information.
7. Maintain shipping and delivery details.
8. Provide order history and business reports.

2.3 Scope

In-Scope:

- Customer registration and management
- Seller management
- Product and category management
- Order creation and cancellation
- Payment management
- Shipping and order tracking
- Inventory-related information
- Order history
- Report generation

Out-of-Scope:

- AI-based recommendations
- Advanced fraud detection
- Real-time GPS tracking
- Voice-based ordering
- Cryptocurrency payments
- Advanced marketing automation

3. Stakeholders and User Roles

Stakeholder	Responsibilities
Customer	Register, browse products, place orders, make payments, track orders, view history
Seller	Add and update products, manage availability, view received orders
Administrator	Manage customers, sellers, products, orders, payments and reports
Delivery Personnel	Receive delivery details, update delivery status and confirm delivery
Business Manager	Analyze sales, revenue, orders, customers and seller performance

These stakeholders interact with or are affected by the system.

4. Business and Functional Requirements

4.1 Business Requirements

ID	Business Requirement
BR-01	Maintain complete customer information.
BR-02	Maintain seller information and seller-product relationships.
BR-03	Store product details, prices and availability.
BR-04	Support creation and management of customer orders.
BR-05	Allow multiple products in one order.
BR-06	Store payment information associated with orders.
BR-07	Maintain shipping and delivery information.

ID	Business Requirement
BR-08	Allow authorized users to track order status.
BR-09	Generate order, product, customer and sales reports.

4.2 Functional Requirements

FR-01: Customer Management
The system shall allow customers to register and authorized users to view, update and deactivate customer information.

FR-02: Product Management
Authorized sellers or administrators shall be able to add, update, remove and view products and their availability.

FR-03: Order Creation
Customers shall be able to create an order by selecting one or more available products.

FR-04: Order Details
Each order shall store the products selected, quantity, unit price and subtotal.

FR-05: Order Modification and Cancellation
Eligible orders can be modified or cancelled according to their status.

FR-06: Payment Management
The system shall store payment method, amount, date and payment status.

FR-07: Shipping Management
The system shall maintain shipping address, courier, tracking information and delivery status.

FR-08: Order Tracking
Customers and authorized users shall be able to view the current order status.

FR-09: Order History
Customers shall be able to view their previous orders.

FR-10: Reporting
Administrators and managers shall be able to generate business reports.

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5. System Modules and Order Workflow

5.1 System Modules

Customer Module

- Registration and login
- Profile management
- Address management
- Order history

Seller Module

- Seller information management
- Product management
- Product availability
- Order monitoring

Product Module

- Add, update and delete products
- Product search
- Availability checking

Order Module

- Create order
- Add products
- Update quantity
- Cancel order
- View order details

Payment Module

- Record payment
- Verify payment status
- Maintain transaction information
- Record refunds

Shipping Module

- Create shipment
- Assign delivery
- Track shipment
- Update delivery status

Reporting Module

- Daily orders
- Monthly sales
- Product sales
- Seller sales
- Payment reports
- Cancelled orders

5.2 Order Processing Workflow

Customer Registration → Product Browsing → Product Selection → Order Creation → Order Details → Amount Calculation → Payment → Payment Verification → Order Confirmation → Shipping → Delivery → Order Completion

For each ordered product:

Subtotal = Quantity × Unit Price

Total Amount = Sum of all Subtotals

Order Status

Pending → Confirmed → Processing → Shipped → Out for Delivery → Delivered

Other possible statuses are **Cancelled** and **Returned**.

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6. Data Requirements and Entities

6.1 Major Data Requirements

Entity	**Important Attributes**
Customer	Customer_ID, Name, Email, Phone, Address
Seller	Seller_ID, Name, Email, Phone, Address

Entity	Important Attributes
Product	Product_ID, Name, Category_ID, Seller_ID, Price, Stock
Category	Category_ID, Category_Name
Order	Order_ID, Customer_ID, Order_Date, Status, Total_Amount
Order_Details	Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal
Payment	Payment_ID, Order_ID, Method, Amount, Date, Status
Shipping	Shipping_ID, Order_ID, Address, Tracking_Number, Status

6.2 Primary Entities

The major entities are:

1. Customer
2. Seller
3. Product
4. Category
5. Order
6. Order_Details
7. Payment
8. Shipping

Each major entity should have a unique identifier such as **Customer_ID, Product_ID, Order_ID, Payment_ID, and Shipping_ID**.

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7. Relationships and Business Rules

7.1 Entity Relationships

Relationship	Cardinality
Customer → Order	1 : M
Seller → Product	1 : M

Relationship	Cardinality
Category → Product	1 : M
Order → Order_Details	1 : M
Product → Order_Details	1 : M
Order → Payment	1 : 1 / 1 : M
Order → Shipping	1 : 1

A customer can place multiple orders, while an order can contain multiple order-detail records. A seller can sell multiple products.

7.2 Business Rules

1. Every customer must have a unique Customer ID.
2. Every seller must have a unique Seller ID.
3. Every product must have a unique Product ID.
4. Every order must belong to a valid customer.
5. An order must contain at least one product.
6. Ordered quantity must be greater than zero.
7. Product price cannot be negative.
8. Order total must be calculated from ordered products.
9. Only available products can normally be ordered.
10. Payment must be associated with a valid order.
11. Every product must be associated with a valid seller.
12. Cancelled orders must not be treated as completed sales.

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8. Non-Functional and Security Requirements

8.1 Non-Functional Requirements

Performance

The system should provide quick database responses. Primary keys, indexes, normalization and optimized SQL queries should be used.

Reliability

The system should maintain accurate transaction information and prevent accidental data loss.

Availability

The database should remain available whenever the e-commerce application is operating.

Scalability

The system should support increasing numbers of customers, products, sellers and orders.

Maintainability

The database structure should be easy to understand, modify and maintain.

Usability

The system should provide a simple and understandable interface.

8.2 Security Requirements

- Users must authenticate before accessing protected functions.
- Different users must have different permissions.
- Customers can access their own orders.
- Sellers can manage their own products.
- Administrators can manage system-wide information.
- Sensitive information must be protected.
- Database access must be restricted to authorized users.
- Regular backups should be maintained.

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9. Validation, Inputs, Outputs and Reports

9.1 Data Validation

Customer: Name cannot be empty, email must be valid, phone number must contain valid digits, and Customer ID must be unique.

Product: Product name is required, price cannot be negative, and stock quantity cannot be negative.

Order: Customer and product must exist, and quantity must be greater than zero.

Payment: Payment amount and Order ID must be valid, and payment status must contain an accepted value.

9.2 Input Requirements

- Customer details and login credentials
- Product name, category, price, quantity and seller
- Customer ID, Product ID, quantity and shipping address
- Payment method, amount and transaction reference

9.3 Output Requirements

Customer Outputs:

- Product list
- Order history
- Order status
- Payment status

Administrator Outputs:

- Customer report
- Product report
- Order report
- Sales report
- Payment report
- Seller report

Seller Outputs:

- Product list

- Available stock
- Customer orders
- Sales information

9.4 Important Reports

- Customer Order History
- Product Sales Report
- Seller Sales Report
- Daily Order Report
- Monthly Sales Report
- Cancelled Order Report
- Payment Report

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10. Acceptance Criteria, Future Enhancements and Conclusion

10.1 Acceptance Criteria

The system will be considered successful when:

1. Customers and sellers can be stored successfully.
2. Products can be associated with sellers.
3. Orders can be created successfully.
4. Multiple products can be added to an order.
5. Order totals are calculated correctly.
6. Payment details can be recorded.
7. Order status can be updated and tracked.
8. Customers can view their order history.
9. Administrators can generate required reports.
10. Invalid data is rejected and data integrity is maintained.

10.2 Future Enhancements

Future versions may include:

- AI-based product recommendations

- Multiple payment gateways
- Real-time delivery tracking
- Coupon and discount management
- Product reviews and ratings
- Automated notifications
- Advanced business analytics

10.3 Conclusion

The **E-Commerce Order Management System** provides a structured solution for managing online business transactions. The requirement analysis identifies the major business needs including customer, seller, product, order, payment, shipping, tracking and reporting.

The system uses a relational database with appropriate entities, relationships, primary keys, foreign keys, validation, normalization and security. This document provides the foundation for further database development, including ER diagram creation, relational schema design, table creation, normalization and SQL implementation.